

Original Solution by: Edmund Bertschinger
Typeset by: Swati Ravi

Problem 1: Central Potential

(a) Period of a circular orbit of radius r

Integrals of motion for a central potential $V(r)$: Angular momentum:

$$L = rV_t = r^2\dot{\phi} \quad (V_t = \text{tangential velocity})$$

Energy per unit mass:

$$E = \frac{1}{2} (\dot{r}^2 + V_t^2) + V(r) = \frac{1}{2} \dot{r}^2 + V_{\text{eff}}(r)$$

Where:

$$V_{\text{eff}}(r) = V(r) + \frac{L^2}{2r^2}$$

For a circular orbit ($\dot{r} = 0$):

$$\frac{dV_{\text{eff}}}{dr} = 0 \quad \rightarrow \quad \frac{dV}{dr} - \frac{L^2}{r^3} = 0 \quad \rightarrow \quad \frac{dV}{dr} = \frac{L^2}{r^3} = r\dot{\phi}^2$$

$$\dot{\phi} = \omega_{\phi} = \frac{L}{r^2} = \left(\frac{1}{r} \frac{dV}{dr} \right)^{1/2} \quad \rightarrow \quad P_{\phi} = \frac{2\pi}{\omega_{\phi}} = 2\pi \left(\frac{1}{r} \frac{dV}{dr} \right)^{-1/2}$$

(b) Slightly noncircular orbit:

$$r(t) = r_0 + \epsilon(t) \quad \text{with} \quad \left(\frac{dV_{\text{eff}}}{dr} \right)_{r_0} = 0 \quad \text{and} \quad \epsilon^2 \ll r_0^2.$$

Energy per unit mass:

$$E = \frac{1}{2} \dot{\epsilon}^2 + V_{\text{eff}}(r_0 + \epsilon).$$

Taylor expand V_{eff} :

$$V_{\text{eff}}(r_0 + \epsilon) = V_{\text{eff}}(r_0) + \left(\frac{dV_{\text{eff}}}{dr} \right)_{r_0} \epsilon + \frac{1}{2} \left(\frac{d^2V_{\text{eff}}}{dr^2} \right)_{r_0} \epsilon^2 + O(\epsilon^3).$$

Substituting into the energy equation:

$$\Rightarrow E - V_{\text{eff}}(r_0) = \frac{1}{2} \dot{\epsilon}^2 + \frac{1}{2} \omega_r^2 \epsilon^2 + O(\epsilon^3) = \text{constant},$$

where:

$$\omega_r = \left(\frac{d^2V_{\text{eff}}}{dr^2} \right)_{r_0}^{1/2}.$$

This is the simple harmonic oscillator equation, and its general solution is given by:

$$\epsilon(t) = \frac{\sqrt{2(E - E_0)}}{\omega_r} \cos[\omega_r(t - t_0)] \quad \text{where} \quad E_0 = V_{\text{eff}}(r_0); \quad t_0 \text{ is an arbitrary constant.}$$

Now, we write ω_r in terms of $V(r)$ instead of $V_{\text{eff}}(r)$:

$$\omega_r^2 = \frac{d^2 V_{\text{eff}}}{dr^2} = \frac{d^2 V}{dr^2} + \frac{3L^2}{r^4} = \frac{d^2 V}{dr^2} + 3\omega_\phi^2,$$

where:

$$\omega_r = \left[\frac{1}{r^3} \frac{d}{dr} \left(r^3 \frac{dV}{dr} \right) \right]_{r_0}^{1/2}.$$

Radial period P_r :

$$P_r = \frac{2\pi}{\omega_r}.$$

c) Stable circular orbits for the Yukawa potential:

The potential is given by:

$$V(r) = -\frac{GM}{r} e^{-kr} \quad \dots (1)$$

Stability is determined by the sign of ω_r^2 :

$$\omega_r^2 = \frac{1}{r^3} \frac{d}{dr} \left(r^3 \frac{dV}{dr} \right), \quad \omega_r^2 > 0 \text{ for stability.}$$

Given $V(r)$, compute:

$$\begin{aligned} \frac{dV}{dr} &= \frac{GM}{r^2} e^{-kr} (kr + 1), \\ r^3 \frac{dV}{dr} &= GM e^{-kr} (kr^2 + r). \end{aligned}$$

Now,

$$\frac{d}{dr} \left(r^3 \frac{dV}{dr} \right) = \frac{d}{dr} (GM e^{-kr} (kr^2 + r)).$$

Using the product rule:

$$\frac{d}{dr} (GM e^{-kr} (kr^2 + r)) = GM e^{-kr} [1 + kr - (kr)^2].$$

Thus:

$$\omega_r^2 = \frac{GM}{r^3} e^{-kr} [1 + kr - (kr)^2].$$

For stability:

$$1 + kr - (kr)^2 > 0,$$

or equivalently:

$$(kr)^2 - kr - 1 < 0.$$

Solving the quadratic equation:

$$(kr)^2 - kr - 1 = 0,$$

the roots are:

$$kr = \frac{1 \pm \sqrt{5}}{2}.$$

The condition $1 + kr - (kr)^2 > 0$ will be satisfied for:

$$\frac{1 - \sqrt{5}}{2} < kr < \frac{1 + \sqrt{5}}{2}.$$

Since $r > 0$, we only consider:

$$0 < kr < \frac{1 + \sqrt{5}}{2}.$$

For $r > \frac{1+\sqrt{5}}{2k}$, the term $[1 + kr - (kr)^2] < 0$, and the circular orbits are unstable. Q.E.D.

d) **Bound orbits with positive energy:**

The outermost stable circular orbit is at

$$r_0 = \frac{1 + \sqrt{5}}{2k}$$

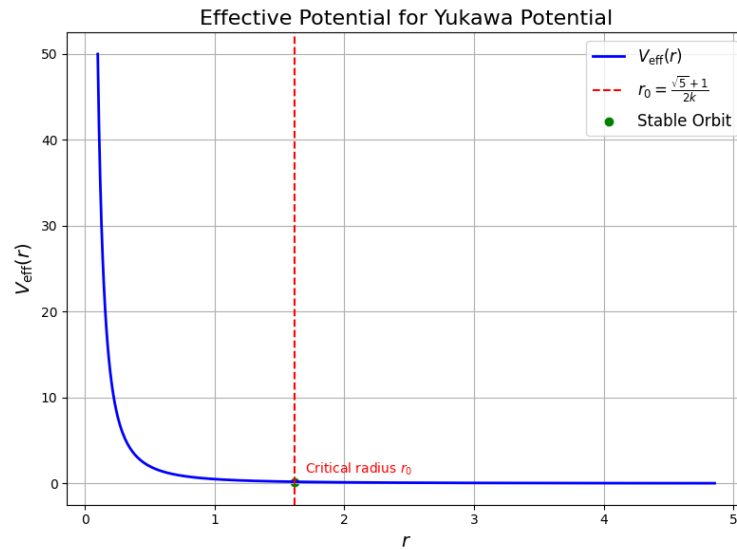
It has energy

$$E = V(r_0) + \frac{1}{2} (r_0 \omega_\phi)^2 = V(r_0) + \frac{1}{2} \left(r \frac{dV}{dr} \right)_{r_0}$$

Substituting:

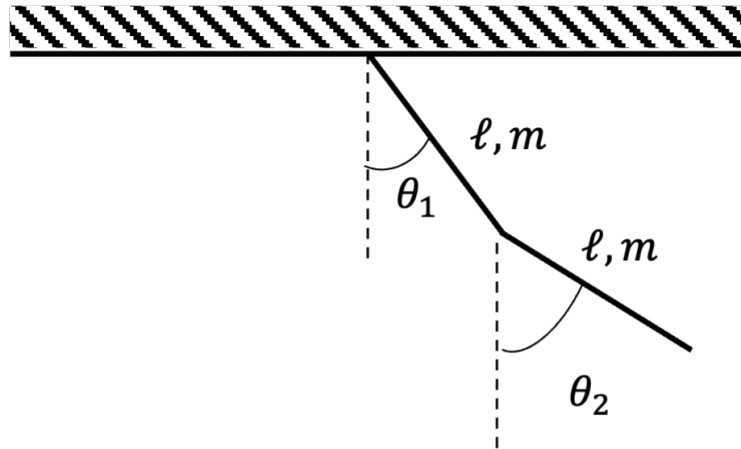
$$E = \frac{GM}{r_0} e^{-kr_0} \left[\frac{1}{2} (kr_0 - 1) \right] = \frac{GM}{r_0} e^{-kr_0} \left(\frac{\sqrt{5} - 1}{4} \right) > 0$$

If r_0 is decreased slightly, the orbit is absolutely stable and $E > 0$. The effective potential for the Yukawa potential has the form shown.



Problem 2: Planar Double Pendulum

a) **Lagrangian:**



$$L = T - V$$

For the first rod:

$$T_1 = \frac{1}{2} \int_0^\ell \left(r \dot{\theta}_1 \right)^2 \left(\frac{m}{\ell} \right) dr = \frac{1}{6} m \ell^2 \dot{\theta}_1^2$$

For the second rod, we must apply a velocity offset. Working in Cartesian coordinates:

$$\begin{aligned} T_2 &= \frac{1}{2} \int_0^\ell \left[\left(\ell \dot{\theta}_1 \cos \theta_1 + r \dot{\theta}_2 \cos \theta_2 \right)^2 + \left(\ell \dot{\theta}_1 \sin \theta_1 + r \dot{\theta}_2 \sin \theta_2 \right)^2 \right] \left(\frac{m}{\ell} \right) dr \\ &= \frac{1}{2} m \ell^2 \left[\dot{\theta}_1^2 + \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) + \frac{1}{3} \dot{\theta}_2^2 \right] \end{aligned}$$

The potential energies are simply mgz , where z is the vertical distance from the ceiling (negative for positions below the ceiling). Since the rods are uniform, we may treat them as if all the mass is concentrated at their center of mass (COM):

$$V_1 = mg \left(-\frac{\ell}{2} \cos \theta_1 \right), \quad V_2 = mg \left[-\ell \cos \theta_1 - \frac{\ell}{2} \cos \theta_2 \right]$$

Expanding the cosines for small angles: $\cos \theta \approx 1 - \frac{\theta^2}{2}$,

$$\cos(\theta_1 - \theta_2) \approx 1 \quad (\text{higher-order terms negligible given angle difference } (\theta_1 - \theta_2))$$

$$\begin{aligned} V_1 + V_2 &= -\frac{mg\ell}{2} \left(1 - \frac{\theta_1^2}{2} \right) - mg\ell \left(1 - \frac{\theta_1^2}{2} \right) - \frac{mg\ell}{2} \left(1 - \frac{\theta_2^2}{2} \right) \\ &= -2mg\ell + mg\ell \left(\frac{3}{4} \theta_1^2 + \frac{1}{4} \theta_2^2 \right) \end{aligned}$$

Thus,

$$L = T_1 + T_2 - V_1 - V_2 = m\ell^2 \left(\frac{2}{3} \dot{\theta}_1^2 + \frac{1}{6} \dot{\theta}_1 \dot{\theta}_2 + \frac{1}{6} \dot{\theta}_2^2 \right) - mg\ell \left(\frac{3}{4} \theta_1^2 + \frac{1}{4} \theta_2^2 \right) + \text{constant}$$

b) Frequencies of small oscillations:

Lagrange's equations:

$$\begin{aligned}\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_i} \right) - \frac{\partial L}{\partial \theta_i} &= 0 \quad i = 1, 2 \\ \frac{\partial L}{\partial \theta_1} &= -\frac{3}{2}mg\ell\theta_1, \quad \frac{\partial L}{\partial \theta_2} = -\frac{1}{2}mg\ell\theta_2 \\ \frac{\partial L}{\partial \dot{\theta}_1} &= \frac{4}{3}m\ell^2\dot{\theta}_1 + \frac{1}{2}m\ell^2\dot{\theta}_2, \quad \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_1} \right) = \frac{4}{3}m\ell^2\ddot{\theta}_1 + \frac{1}{2}m\ell^2\ddot{\theta}_2 \\ \frac{\partial L}{\partial \dot{\theta}_2} &= \frac{1}{3}m\ell^2\dot{\theta}_2 + \frac{1}{2}m\ell^2\dot{\theta}_1, \quad \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_2} \right) = \frac{1}{3}m\ell^2\ddot{\theta}_2 + \frac{1}{2}m\ell^2\ddot{\theta}_1\end{aligned}$$

Substituting these into the equations of motion:

$$\begin{aligned}\frac{4}{3}\ddot{\theta}_1 + \frac{1}{2}\ddot{\theta}_2 + \frac{g}{\ell} \left(\frac{3}{2}\theta_1 \right) &= 0 \\ \frac{1}{2}\ddot{\theta}_1 + \frac{1}{3}\ddot{\theta}_2 + \frac{g}{\ell} \left(\frac{1}{2}\theta_2 \right) &= 0\end{aligned}$$

Assume normal modes:

$$\theta_1 = \hat{\theta}_1 e^{i\omega t}, \quad \theta_2 = \hat{\theta}_2 e^{i\omega t} \quad \rightarrow \quad \ddot{\theta}_1 = -\omega^2 \hat{\theta}_1 e^{i\omega t}, \quad \ddot{\theta}_2 = -\omega^2 \hat{\theta}_2 e^{i\omega t}$$

Substituting into the equations of motion gives:

$$\begin{pmatrix} \frac{4}{3}\omega^2 - \frac{3g}{2\ell} & \frac{\omega^2}{2} \\ \frac{\omega^2}{2} & \frac{1}{3}\omega^2 - \frac{g}{2\ell} \end{pmatrix} \begin{pmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{pmatrix} = 0$$

Let $\frac{g}{\ell} = \omega_0^2$ and $X = \frac{\omega^2}{\omega_0^2}$. The determinant of the square matrix M above must equal 0 for non-trivial solutions:

$$\begin{aligned}\det \begin{pmatrix} \frac{4}{3}X - \frac{3}{2} & \frac{X}{2} \\ \frac{X}{2} & \frac{1}{3}X - \frac{1}{2} \end{pmatrix} &= 0 \\ \left(\frac{4}{3}X - \frac{3}{2} \right) \left(\frac{1}{3}X - \frac{1}{2} \right) - \frac{X^2}{4} &= 0 \\ \frac{4}{9}X^2 - \frac{1}{4}X^2 - \frac{1}{2}X - \frac{2}{3}X + \frac{3}{4} + \frac{3}{4} &= 0 \\ \frac{7}{36}X^2 - \frac{7}{6}X + \frac{3}{4} &= 0\end{aligned}$$

Solving the quadratic equation:

$$X = 3 \pm \frac{6}{\sqrt{7}} = \frac{\omega^2 \ell}{g}$$

The normal mode frequencies are:

$$\omega = \sqrt{\frac{gX}{\ell}}$$

(c) Sketch the normal modes:

Find the eigenvectors.

- **First case:** $\omega^2 = \left(3 - \frac{6}{\sqrt{7}}\right) \frac{g}{\ell}$ (low frequency mode)
 or $X = 3 - \frac{6}{\sqrt{7}}$

$$\hat{\theta}_2 = \left(\frac{3}{X} - \frac{8}{3}\right) \hat{\theta}_1 = \frac{1}{3}(2\sqrt{7} - 1)\hat{\theta}_1$$

RHS: $\frac{1}{3}(2\sqrt{7} - 1) > 0$ and real $\implies$ **The two pendula are in phase.**

- **Second case:** $\omega^2 = \left(3 + \frac{6}{\sqrt{7}}\right) \frac{g}{\ell}$ (high frequency mode)
 or $X = 3 + \frac{6}{\sqrt{7}}$

$$\hat{\theta}_2 = \left(\frac{3}{X} - \frac{8}{3}\right) \hat{\theta}_1 = \frac{1}{3}(-2\sqrt{7} - 1)\hat{\theta}_1$$

RHS: $\frac{1}{3}(-2\sqrt{7} - 1) < 0$ and real $\implies$ **The two pendula are π out of phase.**

